



"WOULD YOU CALL THIS A PANDEMIC? THE FLU IS PAST ITS PRIME. IN FACT, IT IS PLATEAUING. IS THIS THE RIGHT TIME TO INTRODUCE A VACCINE?"

Dr M.G. DEO, Pathologist and Former Chief, Cancer Research Institute

global majors. The US has the highest infection rate (47 million), while Australia saw the highest number per capita (462 per million). It's quite clear now that there are likely to be far fewer deaths from H1N1 than was initially believed. So why does one need to get vaccinated?

Don't worry, the Government probably doesn't have you on the radar. "We will give it to high-risk groups," says Dr V.M. Katoch, director-general of the Indian Council of Medical Research. Apart from healthcare professionals, pregnant women, emergency workers, etc, the virus is expected to lose out through collective natural immunity. "A com-

munity develops immunity when more and more people get infected and start developing resistance to the pathogen," he says. He mentions a study done on Panchgani school-children in Pune: "About 60 per cent of children developed antibodies within a week. Less than 5 per cent got the clinical disease. And all responded to the drug."

A sound strategy? To Dr Randeep Guleria, professor of medicine, AIIMS, the main cause for worry lies elsewhere. "Genetic mutations arise continuously in influenza viruses—what we call 'drifts'—as both seasonal and pandemic H1N1 flu viruses circulate among the population." The two

strains could recombine into a more virulent and aggressive version, causing widespread illness and death. The more people get vaccinated on time, the less the chances of a severe 'second wave'.

But India has, in a way, missed the bus—precious time, that is. To be effective, a vaccine has to be given in two doses before the start of the official 'flu season' (in this case, at the start of the so-called second wave around November-December). No wonder, mass vaccinations started in much of the developed West—the US, the UK, France, Canada, Australia, Germany—and even China since October. "But India expects to vaccinate healthcare and emergency workers from around February, when the projected pandemic would in fact be waning," says Guleria. There will be a question mark on its utility.

Listen to what experts say on the effectiveness of vaccines. "Vaccination may be the crucial line of defence in most countries, but every flu vaccine is not effective," says Katoch. Western countries have a background of regularly using seasonal flu vaccines. It's easier for them to come up with a new flu vaccine using a new strain. But influenza has never been on India's radar. "We had to start from scratch, needed more start-up time, had to fast-track the new vaccine," he adds.

"But what if they don't work or have undesirable effects," asks Deo. He recalls the Swine Flu Fiasco of 1976—a projected pandemic in the US that actually never happened. The new vaccine was administered to 40 million people and resulted in severe pulmonary complications and Guillian Barre Syndrome (an autoimmune disorder). "Because of the unacceptable side effects, the vaccine was withdrawn and the US government paid millions as compensation."

Epidemiology, economics and ethics form the three pillars of public health. To Dr T. Jacob John, a distinguished force in infectious diseases in India and the former head of clinical virology at Christian Medical College, Vellore, the pandemic management

The flu in India

28,251 cases so far in India, far less than the global majors

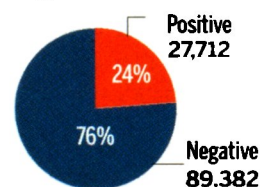
1,136 death toll in India, double the global average

45% of deaths due to co-morbidity: diabetes and other chronic ailments etc

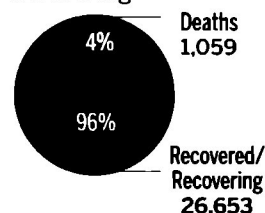
50% in the 16-45 make it the worst-affected age group

75% of those who died came from urban areas

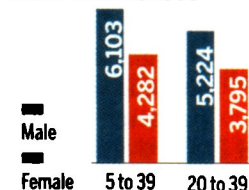
1,17,094 people tested till January 2010. 77% tested negative



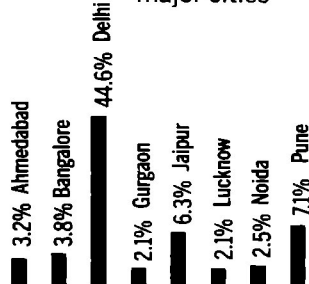
96% cases recovered/recovering



Pattern of total lab confirmed cases



71% cases from major cities



The Vaccine Race

Zydus Cadila is working on egg-based, inactivated vaccine

The Serum Institute is into the only live-virus vaccine to be inhaled through the nose

Bharat Biotech developing the first cell culture-based vaccine

Panacea Biotec is working on conventional egg-based technology

Sources: MoHFW, GOI, Indian J Med Res